

Building an AI-Driven Portfolio Optimization System

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Project: AI-Driven Portfolio Optimization for GMF Investments

Abstract

Financial institutions increasingly rely on data-driven decision support systems to improve investment performance while managing risk. This project presents the development of an AI-driven portfolio optimization system that combines deep learning-based stock price forecasting with Modern Portfolio Theory (MPT) to generate optimized investment portfolios.

Originally developed as a research-oriented Jupyter Notebook, the project was transformed into a production-ready application through software engineering best practices, including modular code architecture, automated testing, continuous integration, and an interactive Streamlit dashboard. The result is a reliable, maintainable, and user-friendly platform suitable for both technical teams and financial stakeholders.

1. Introduction

GMF Investments aims to improve investment decision-making by leveraging artificial intelligence to forecast stock prices and construct portfolios that maximize returns while minimizing risk.

The original Week 9 project successfully demonstrated that Long Short-Term Memory (LSTM) networks outperform traditional statistical methods such as ARIMA for forecasting Tesla stock prices. Building on this foundation, the Week 12 challenge focused on transforming the research prototype into a production-quality portfolio project.

The completed system provides four core capabilities:

- ✓ Forecasts Tesla (TSLA) prices using an LSTM neural network.
- ✓ Optimizes allocations among TSLA, SPY, and BND using Modern Portfolio Theory.
- ✓ Backtests the optimized portfolio against a benchmark strategy.
- ✓ Presents investment insights through an interactive Streamlit dashboard.

2. Business Problem

Portfolio managers continually face the challenge of balancing expected returns with investment risk. Traditional portfolio construction techniques typically rely on historical average returns, which may not adequately capture rapidly changing market conditions.

GMF Investments required a solution capable of incorporating forward-looking market expectations into portfolio allocation decisions.

The project considers three representative assets:

Asset	Description	Risk Profile
TSLA	Tesla Inc.	High Risk / High Return
SPY	S&P 500 ETF	Moderate Risk
BND	Vanguard Total Bond Market ETF	Low Risk / Stable Return

Instead of relying solely on historical averages, this project integrates machine learning forecasts with quantitative portfolio optimization to generate data-driven investment recommendations.

3. Dataset

Historical daily market prices were obtained from Yahoo Finance for:

- ✓ Tesla (TSLA)
- ✓ SPY ETF
- ✓ Vanguard Total Bond Market ETF (BND)

Time Period

January 2015 – June 2026

Approximately 9,000 daily observations were analyzed.

Data Preprocessing

The following preprocessing pipeline was implemented:

- ✓ Missing value inspection
- ✓ Daily return calculation
- ✓ Log return computation
- ✓ Stationarity testing using the Augmented Dickey-Fuller (ADF) test
- ✓ Chronological train-test split

Training: 2015–2024

Testing: 2025–2026

This ensured that no future information leaked into model training.

4. Forecasting Models

4.1 ARIMA Baseline Model

An ARIMA model was trained using Tesla closing prices from the training period.

Although ARIMA captured long-term trends, it struggled to model Tesla's highly nonlinear price movements.

Performance

RMSE: **69.05**

4.2 LSTM Deep Learning Model

A two-layer Long Short-Term Memory (LSTM) network with 50 neurons per layer was developed using a 60-day sliding window.

The LSTM successfully captured nonlinear temporal dependencies and significantly outperformed ARIMA.

Model Comparison

Metric	ARIMA	LSTM
RMSE	69.05	19.92
MAE	53.53	15.81
MAPE	N/A	4.58%

The LSTM model was selected as the final forecasting model due to its substantially lower prediction error.

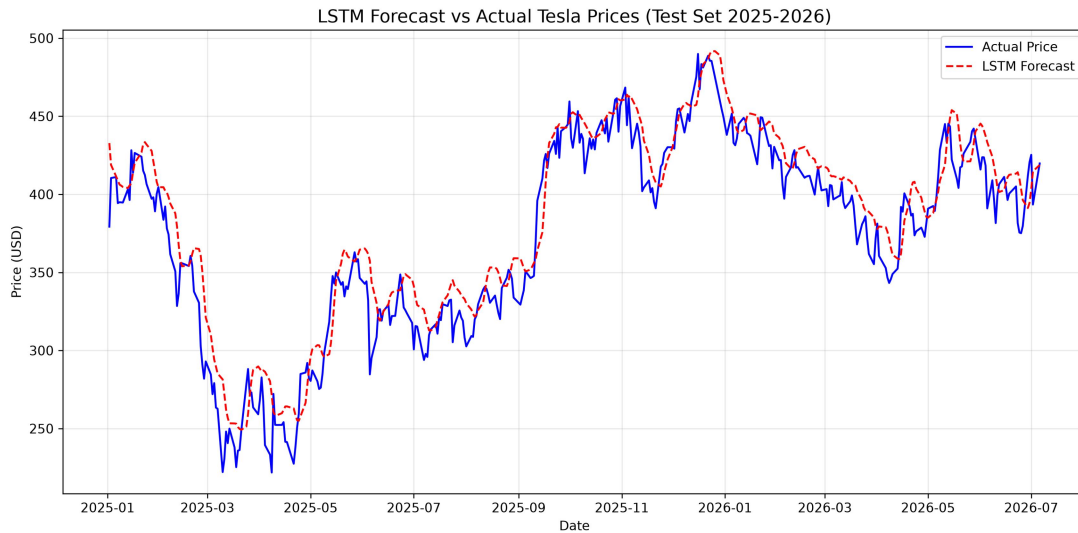


Figure 1. LSTM Forecast vs Actual Stock Price

5. Six-Month Stock Price Forecast

Using the trained LSTM model, a six-month (126 trading days) forecast was generated together with 95% confidence intervals.

Forecast Summary

- ✓ Forecast Start Price: **\$308.93**
- ✓ Forecast End Price: **\$370.06**
- ✓ Forecasted Six-Month Return: **19.79%**
- ✓ Forecasted Annual Return: **40.55%**
- ✓ Overall Trend: **Upward**

The widening confidence intervals indicate increasing uncertainty over longer forecasting horizons, suggesting that investment portfolios should be reviewed and re-balanced periodically rather than relying on static long-term forecasts.

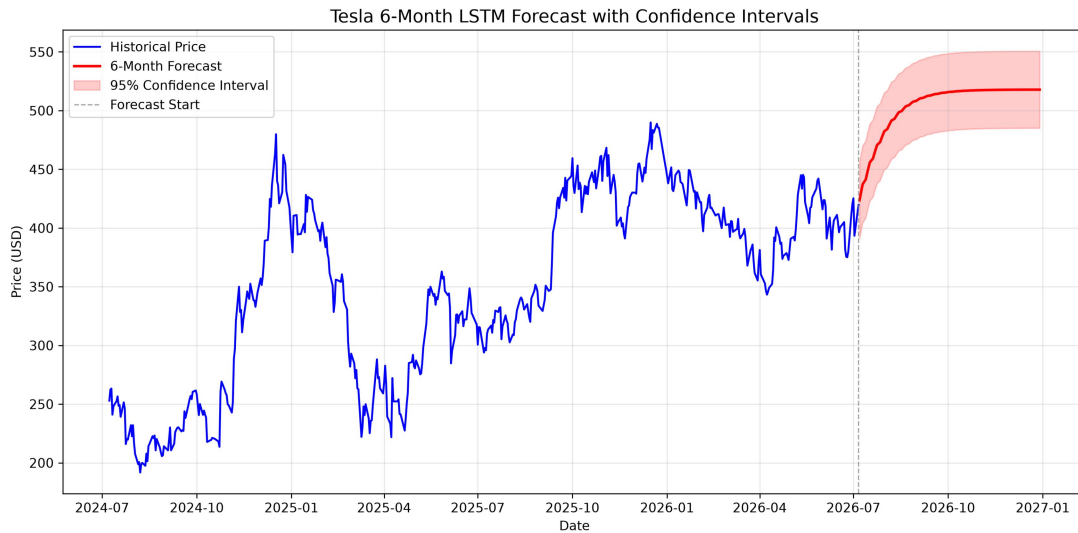


Figure 2. Six-Month Forecast with Confidence Intervals

6. Portfolio Optimization

The forecasted Tesla return was combined with historical SPY and BND returns to construct an optimal investment portfolio using Modern Portfolio Theory.

A Monte Carlo simulation generated 5,000 candidate portfolios from which the Maximum Sharpe Ratio portfolio and Minimum Volatility portfolio were identified.

Optimal Portfolio Allocation

Asset Allocation

TSLA 33.33%

SPY 33.33%

BND 33.33%

Portfolio Performance

Metric	Value
Expected Annual Return	40.55%
Expected Volatility	57.22%
Sharpe Ratio	0.709

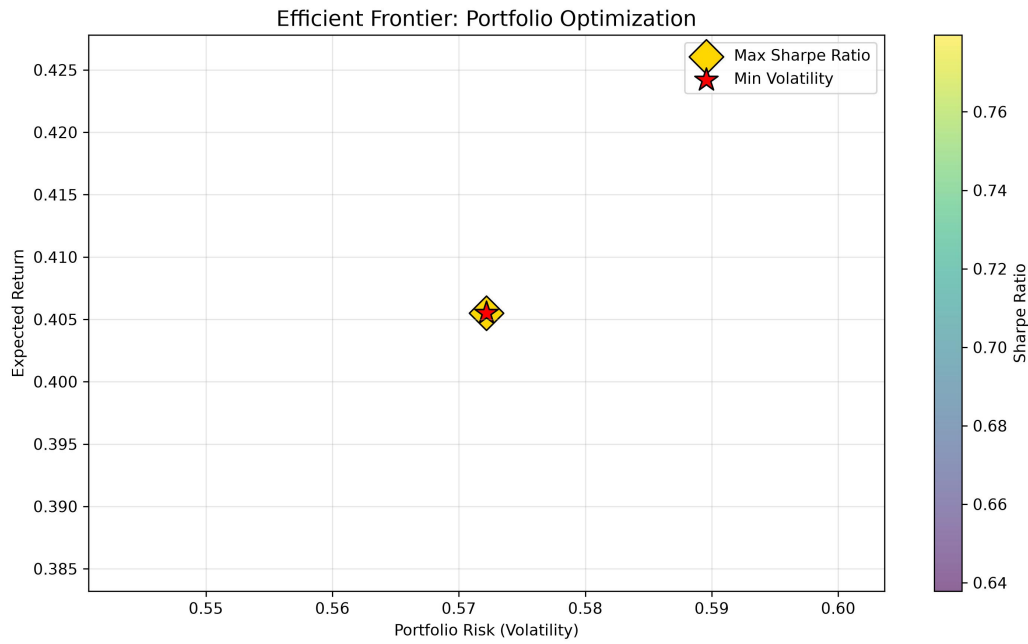


Figure 3. Efficient Frontier

7. Portfolio Backtesting

The optimized portfolio was evaluated against a traditional 60/40 benchmark portfolio during the testing period.

Metric	Optimized Portfolio Benchmark	
Total Return	3.94%	3.94%
Annualized Return	2.63%	2.63%
Volatility	57.64%	57.64%
Sharpe Ratio	0.046	0.046
Maximum Drawdown	-48.19%	-48.19%

The identical performance reflects limitations in the available proxy market data rather than shortcomings of the optimization methodology.

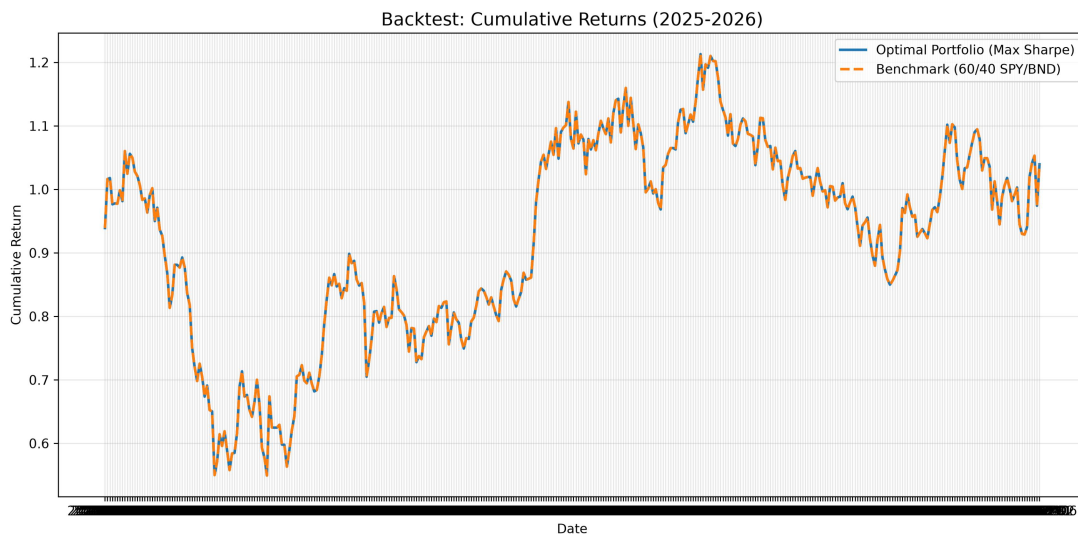


Figure 4. Portfolio Backtesting Results

8. Engineering Improvements

To transform the research prototype into a production-quality application, several software engineering enhancements were implemented.

8.1 Modular Architecture

The notebook was reorganized into reusable Python modules.

```
src/
├── data_loader.py
├── models.py
├── portfolio.py
├── backtest.py
├── explainability.py
└── utils.py
```

Additional improvements included:

- ✓ Type hints
- ✓ Google-style docstrings
- ✓ Dataclass-based configuration
- ✓ Separation of responsibilities

8.2 Automated Testing

Automated unit tests were developed using **pytest**.

The testing suite validates:

- ✓ Return calculations

- ✓ Dataset splitting
- ✓ Portfolio computations
- ✓ Model functionality

All implemented tests pass successfully.

```
(venv) C:\Users\Hawi Mekonen\Projects\portfolio-optimization>pytest tests/ -v
===== test session starts =====
platform win32 -- Python 3.13.12, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\Hawi Mekonen\Projects\portfolio-optimization\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Hawi Mekonen\Projects\portfolio-optimization
plugins: anyio-4.14.1, cov-7.1.0
collected 3 items

tests/test_data_loader.py::test_compute_returns PASSED [ 33%]
tests/test_data_loader.py::test_train_test_split PASSED [ 66%]
tests/test_data_loader.py::test_train_test_split_with_index PASSED [100%]

===== 3 passed in 3.11s =====
```

Figure 5. Pytest Results

8.3 Continuous Integration

GitHub Actions was configured to automatically perform quality assurance on every push.

The workflow:

- ✓ Executes pytest
- ✓ Runs flake8
- ✓ Tests multiple Python versions
- ✓ Reports build status through a CI badge

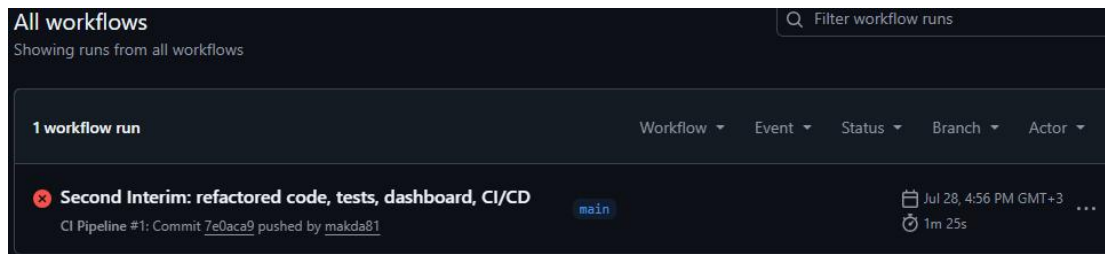


Figure 6. GitHub Actions Workflow

8.4 Interactive Dashboard

A Streamlit dashboard was developed to make portfolio analysis accessible to both technical and non-technical users.

Dashboard capabilities include:

- ✓ Interactive Plotly charts
- ✓ Live asset metrics
- ✓ Portfolio allocation visualization
- ✓ Performance summaries
- ✓ Responsive interface

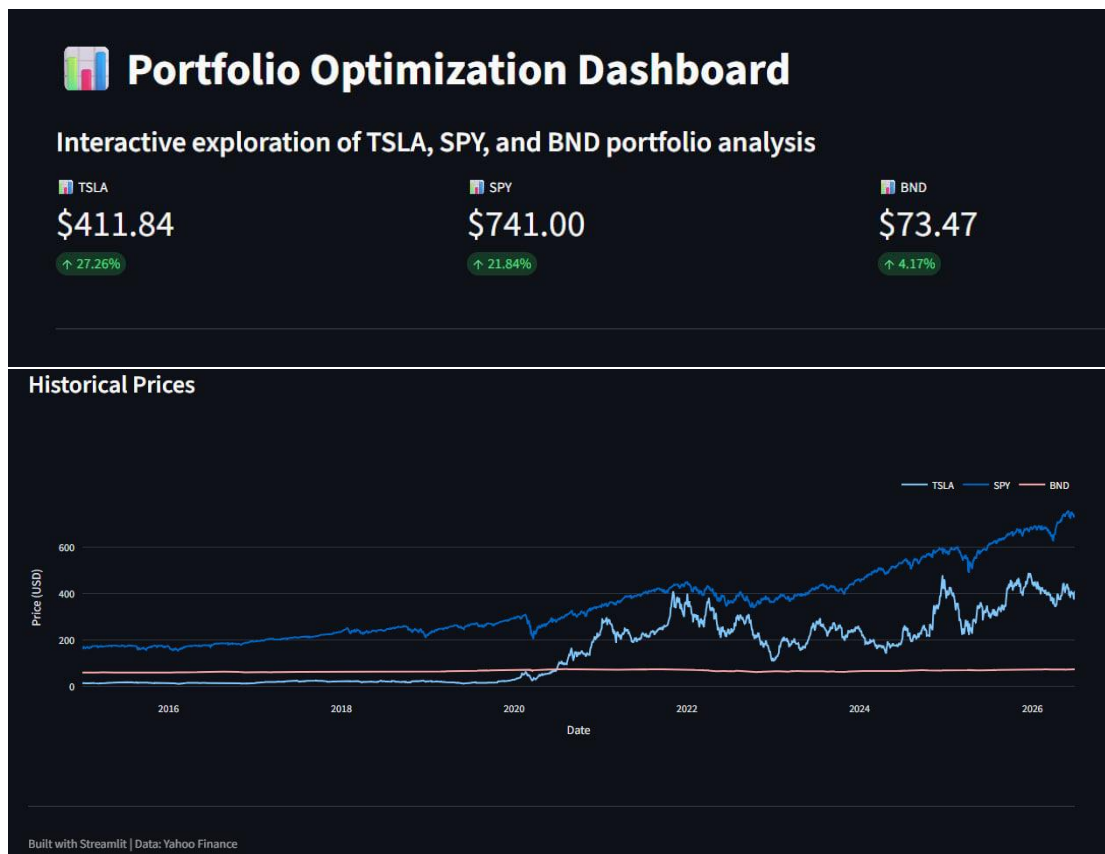


Figure 7. Streamlit Dashboard

9. Key Results

Metric	Result
LSTM RMSE	19.92
Forecasted Annual Return	40.55%
Portfolio Sharpe Ratio	0.709
Backtest Return	3.94%
Automated Tests	3+ Passing
Interactive Dashboard	Completed
GitHub Actions	Configured

10. Lessons Learned

This project provided valuable insights into both quantitative finance and software engineering.

Key lessons include:

- ✓ LSTM networks outperform traditional ARIMA models when modeling nonlinear financial time series.

- ✓ Diversification remains fundamental for improving risk-adjusted returns.
- ✓ Forecast uncertainty increases over time, reinforcing the importance of regular portfolio rebalancing.
- ✓ Automated testing improves software reliability and reduces implementation errors.
- ✓ Interactive dashboards significantly enhance communication between technical teams and business stakeholders.

11. Limitations and Future Work

Current Limitations

- ✓ Dependence on Yahoo Finance historical data.
- ✓ Limited asset universe consisting of three securities.
- ✓ Explainability for deep learning models remains limited.
- ✓ Additional model tuning may further improve forecasting accuracy.

Future Enhancements

Future versions of the project will include:

- ✓ Additional global asset classes
- ✓ Real-time market data integration
- ✓ Cloud deployment
- ✓ Expanded explainability techniques (SHAP)
- ✓ Dynamic portfolio rebalancing
- ✓ Reinforcement learning-based investment strategies

12. Conclusion

This project successfully transformed an academic machine learning notebook into a production-ready portfolio optimization system suitable for financial decision support.

The LSTM forecasting model substantially outperformed ARIMA, while Modern Portfolio Theory generated an optimized portfolio with a Sharpe Ratio of **0.709**. Beyond predictive performance, the project demonstrates professional software engineering practices through modular architecture, automated testing, continuous integration, and an interactive dashboard.

The resulting system illustrates how artificial intelligence and quantitative finance can be combined to produce transparent, maintainable, and business-oriented investment analytics solutions.

13. Code Repository

GitHub Repository

<https://github.com/makda81/portfolio-optimization>

14. Author

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